

Day -6 : Introduction to Network Device

1. Network Devices: Functions & Typical Ports :

Network Device	Primary Function in Home Network	Typical Physical Ports Used
Switch	Connects multiple wired local devices (PCs, game consoles, TVs) together within the same network so they can share data.	Ethernet (RJ-45) ports (usually 4 to 8 ports on home units).
Router	Routes data between your home local network (LAN) and the internet (WAN), assigning local IP addresses.	WAN/Internet port (RJ-45) , LAN ports (RJ-45) , and a Power port.
Hub	Passes incoming data to <i>all</i> connected devices unconditionally (outdated, mostly replaced by switches).	Ethernet (RJ-45) ports.
Firewall	Filters incoming and outgoing traffic based on security rules to block malicious access.	Integrated into the WAN & LAN ports of home routers (virtual/software boundaries) or dedicated Ethernet interfaces.
Access Point (AP)	Broadcasts wireless Wi-Fi signals so phones and laptops can connect wirelessly to the wired network.	Ethernet (RJ-45) port (often supporting PoE - Power over Ethernet) and Power port.

2. Physical Port Identification Guide :

When looking at a physical network device or hardware diagram, you can identify and label the device ports using these distinct features:

- **Wireless Router / Access Point:**
 - **Antennas:** External adjustable rods (or internal housing) broadcasting 2.4GHz / 5GHz / 6GHz Wi-Fi signals.
 - **Ethernet Ports (RJ-45):** Wide rectangular jacks with clip slots. Color-coded into **WAN/Internet** (usually yellow or blue, single port) and **LAN** (usually grey/black, grouped in 4s).
 - **Power Port:** Small round barrel jack (DC input) or standard C14 plug power connector.
- **Network Switch:**
 - **Ethernet Ports (RJ-45):** Rows of uniform rectangular jacks with built-in green/amber activity LED lights.
 - **Power Port:** Located on the rear side (DC barrel jack or standard 3-prong AC power inlet).
- **Enterprise Router / Firewall:**
 - **Console Port:** An RJ-45 or USB-C port labeled "**Console**" (often light blue), used directly for initial command-line configuration.
 - **Ethernet/SFP Ports:** Labeled WAN/LAN or numbered interfaces (*GE0/0*, *GE0/1*).
 - **Power Port:** Main AC inlet on the rear panel, accompanied by a dedicated power switch.

3. Daily App Network Path: Instagram :

When you open **Instagram** to load reels and photos on your phone, two main devices carry your data:

- **Access Point (AP):** Your phone sends the request wirelessly over Wi-Fi to the Access Point, which converts those radio waves into wired network signals.
- **Router & Firewall:** The built-in Firewall checks that your outgoing request is safe, and the Router directs your request out to Instagram's web servers on the internet to fetch the media files back to your screen.

4. College Wi-Fi Issues & Upgrade Solution :

- **Possible Cause: Access Point (AP) Overload & Congestion.**
Standard consumer Access Points can typically handle only 30–50 connected devices smoothly. In a college hall, hundreds of students connect smartphones, laptops, and tablets to the same AP simultaneously. This overwhelms its antenna capacity and bandwidth allocation, causing high latency, slow speeds, and random disconnections.
- **How Upgrading Helps:** Upgrading to an enterprise-grade **Wi-Fi 6** or **Wi-Fi 6E Access Point** with high user capacity (MU-MIMO technology) allows the AP to communicate with dozens of devices at the exact same instant instead of making them queue up. This eliminates the bottleneck, expands coverage area, and keeps connection drops from happening even during crowded campus hours.

[How To Set Up An Access Point \(AP\) On Your Home Router](#)

This video provides a practical step-by-step visual demonstration of how an Access Point connects to network hardware like routers and switches to improve Wi-Fi coverage.